Time: 1 hr. Paper Code: DSC-502	B. Tech. CSE (Data Science V Semester) 1st Sessional Test 2026-27 Data Analytics	Max. Marks: 12

Note: Attempt All Questions. Make suitable assumptions wherever necessary.

Q1. CO1	**(a)** Calculate Pearson's correlation coefficient for the following data representing advertising expenditure and sales: X=[10,20,30,40,50] and Y=[25,35,45,60,70]. Determine whether the variables have a strong, moderate, or weak correlation and state its direction. **(b)** MegaMart recorded daily sales (in ₹ thousands) for 9 days: 45,52,48,60,55,40,58,50, 47 (i) Calculate the mean, variance, and standard deviation of daily sales. (ii) The store manager wants to predict next week's staffing needs. How would a high standard deviation in daily sales influence the decision, compared to relying on the mean alone?	12+ BL
Q2. CO2	How the objectives, statistical measures, and visualization techniques vary across univariate, bivariate, and multivariate analysis? Present the comparison using a well-structured table. Why a Data Analyst must carry out these three stages of analysis in a sequential manner before using the data to train a machine learning model?	14 BL
Q3. CO3	MedInsight Diagnostics, a healthcare analytics company, uses gene-expression data to predict disease risk. As a Data Scientist, you are given a dataset containing over 20,000 gene-expression features for only a few thousand patients. Analyze the *Curse of Dimensionality* in the context of MedInsight's gene-expression dataset. Explain how Principal Component Analysis (PCA) and t-SNE can be used to address challenges associated with high-dimensional data. Justify which technique would be more appropriate at different stages of MedInsight's disease-risk prediction pipeline and support your choice with suitable reasoning.	14 BL